FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL (G01)
LAB-4. ETL Pipeline with Data Quality Validation (Great Expectations)

1. Context

You and your team are working as data engineers at a retail company operating across Latin America. The business team needs a reliable analytics system to answer key questions about sales performance, customer behavior, and product demand across countries.

However, the available transactional data — captured directly from point-of-sale systems — contains multiple quality issues that make it unfit for analytics as-is:

- Missing and malformed values in key fields
- Duplicate transaction records inflating revenue figures
- Inconsistent country name formats across branches
- Negative quantities and prices from data entry errors
- Derived totals that do not match their source fields
- Transaction dates in mixed formats and out-of-range values

Ensuring data quality is therefore essential for trustworthy analytics and competitive advantage.

2. General Objective

Design and implement a complete ETL pipeline that incorporates data validation using Great Expectations, applies appropriate cleaning and transformation strategies, and produces reliable business insights from a dimensional model.

3. Learning Outcomes

By the end of this lab, students will be able to:

- design a modular ETL pipeline following professional project conventions
- implement validation rules using Great Expectations (Expectation Suites, Data Docs)
- identify, classify, and quantify data quality issues across six dimensions
- apply and justify data cleaning strategies per issue type
- distinguish between Cleaning and Transformation as separate pipeline stages
- validate data before and after transformation using the same GE framework
- design and implement a dimensional model (star schema) for a retail scenario
- connect data quality decisions to measurable business outcomes

4. Dataset

The dataset for this lab is provided as a CSV file: retail_etl_dataset.csv.
4.1 Column Description

Column	Description	Expected Type	Expected Format / Range

invoice_id	Unique identifier for each transaction	Integer	Positive integer, unique
customer_id	Identifier of the customer who made purchase	Integer	Positive integer, not null
product	Name of the purchased product	String	From fixed catalog (8 items)
quantity	Number of units purchased	Integer	≥ 1
price	Unit price of the product at time of sale	Float	> 0.0 (USD)
total_revenue	Total amount = quantity × price	Float	> 0.0 (derived field)
country	Country where transaction occurred	String	Title case, from fixed set
invoice_date	Date the transaction took place	Date	YYYY-MM-DD, within 2023

4.2 Known Data Quality Issues

The following problems are present in the raw dataset. Students are expected to discover, quantify, and address all of them:

Dimension	Column(s)	Problem	Approx. Count
Completeness	customer_id, invoice_date	NULL customer_id (as float NaN); null-like strings in invoice_date ('N/A', 'NULL', '')	202 + 15
Uniqueness	invoice_id	Duplicate invoice_id rows — same ID appears multiple times, inflating revenue	1,159 (~23%)
Validity	quantity, price	Negative values in quantity (e.g. -3) and price (e.g. -83.02) — physically impossible	149 + 101
Consistency	country	Same country in multiple formats: 'Colombia', 'colombia', 'CO'; 'Ecuador', 'ecuador'	~1,500
Accuracy	total_revenue	total_revenue ≠ quantity × price in some rows (offset errors, wrong multiplier, sign flip)	148

Timeliness	invoice_date	Future dates (2025–2026); mixed formats: YYYY/MM/DD and DD-MM-YYYY alongside YYYY-MM-DD	30 + 98 + 78

5. Business Objectives

Your ETL pipeline must support the following business goals. Each is referenced in the tasks below.

BO-1 — Ensure Financial Integrity
Guarantee that revenue reports accurately reflect real sales by eliminating duplicates, negative values, and inaccurate totals. This allows management to trust financial KPIs.

BO-2 — Support Strategic Planning with Reliable Trends
Standardize transaction dates and ensure completeness to enable time-based analyses (monthly growth, seasonal peaks). This helps managers optimize inventory and promotions.

BO-3 — Strengthen Product and Regional Insights
Ensure product names and country data are consistent and standardized. This allows teams to identify top-selling products and regional performance with confidence.

BO-4 — Enable Transparent and Defensible Reporting
Deliver clean, validated data with traceable quality decisions. This fosters trust among stakeholders by showing that reported figures are based on high-quality, auditable information.

6. Solution Architecture

Raw Data
↓
Extrac + Data Profiling
↓
Input Validation (GE)
↓
DQ Analysis
↓
Cleaning
↓
Transformation
↓
Post-Transformation Validation (GE)
↓
Dimensional Modeling (Star Schema)
↓
Load (Data Warehouse - local)
↓
Business Analysis

7. Tasks

a. Extract & Profiling

Load the dataset into a Pandas DataFrame and register it as a Great Expectations data source (in-memory Pandas datasource). Then perform systematic profiling to understand the raw data before writing any expectations.

Your profiling script must compute and report:

- Shape (rows, columns), dtypes, memory usage
- Missing value count and percentage per column
- Cardinality of all categorical columns (product, country, invoice_date formats)
- Descriptive statistics for numeric columns (quantity, price, total_revenue): min, max, mean, median, std
- Number of duplicate invoice_id values
- Count of rows where total_revenue ≠ quantity × price (within tolerance ±0.01)
- Distribution of invoice_date formats (YYYY-MM-DD vs YYYY/MM/DD vs DD-MM-YYYY vs other)
- Count of future dates (> 2023-12-31) and null-like date strings

> *Deliverable: profiling summary table (one row per column) in your PDF report.*

b. Input Data Validation (Great Expectations)

Business objectives: BO-1, BO-2, BO-3, BO-4
Build a formal Expectation Suite for the raw input data. The purpose of this suite is to measure and document quality problems — most expectations are expected to FAIL. Failure rates are your quality baseline.
Your suite must include at least one expectation per quality dimension, covering all columns with known issues:

Dimension	Column(s)	Minimum Expectation	Expected Result
Completeness	customer_id, invoice_date	expect_column_values_to_not_be_null	FAIL
Uniqueness	invoice_id	expect_column_values_to_be_unique	FAIL
Validity	quantity	expect_column_values_to_be_between(min=1)	FAIL
Validity	price	expect_column_values_to_be_between(min=0.01)	FAIL
Validity	product	expect_column_values_to_be_in_set(…)	FAIL
Accuracy	total_revenue	Custom: total_revenue == qty × price	FAIL
Consistency	country	expect_column_values_to_be_in_set(…)	FAIL

Timeliness	invoice_date	expect_column_values_to_match_regex(YYYY-MM-DD)	FAIL
Timeliness	invoice_date	expect_column_values_to_be_between('2023-01-01','2023-12-31')	FAIL

Run validation on the raw data and generate a Data Docs report. Document the failure rate for each expectation.

> **Deliverables:** *Expectation Suite JSON file; Data Docs HTML report; failure rate summary table in PDF.*

c. Data Quality Analysis and Policy Proposal

Business objectives: BO-1, BO-2, BO-3, BO-4
Based on the profiling and validation results, produce two artifacts:

c.1 Data Quality Issues Table

Fill in a row for every issue found:

Column	Issue	Example	Dimension	Business Impact
invoice_id	Duplicate IDs	22951 appears 3×	Uniqueness	Revenue double/triple-counted in BO-1 KPIs
customer_id	NULL values	NaN in 202 rows	Completeness	Cannot link sales to customers for BO-3
quantity	Negative values	-3 in row 5421	Validity	Negative units corrupt total_revenue (BO-1)
price				
total_revenue				
country				
invoice_date				
invoice_date				

c.2 Data Quality Policy Proposal

Formalize at least 8 policies. Each must include a plain-English statement, a GE expectation, and a severity level. The first 6 are given as a starting template — you must add at least 2 original policies based on your own findings:

#	Policy Statement	GE Expectation	Severity	Addresses (BO)
P-01	invoice_id must be	expect_column_values_to_be_unique('invoice_id')	**Critical**	BO-1, BO-4

	unique across the entire dataset.			
P-02	quantity must be a positive integer (≥ 1).	expect_column_values_to_be_between('quantity', min_value=1)	**Critical**	BO-1
P-03	price must be greater than zero.			
P-04	total_revenue must equal quantity × price (±0.01 tolerance).			
P-05	country must be one of: {Colombia, Ecuador, Peru, Chile} (title case, no abbreviations).			
P-06	invoice_date must follow YYYY-MM-DD and fall within 2023-01-01 to 2023-12-31.			
P-07	[Student-defined policy 1]			
P-08	[Student-defined policy 2]			

Deliverables: Data Quality Issues table + Policy Proposal table in PDF report.

d. Cleaning

Definition: Cleaning corrects or removes records that are broken or invalid. A record that fails cleaning cannot safely enter the pipeline. Apply cleaning strategies only when needed and justify every decision.

Business objectives: BO-1, BO-2, BO-3

Issue	Cleaning Strategy	Justification	Log Requirement
Duplicate invoice_id	Remove duplicates; keep first occurrence	Duplicate rows inflate revenue (BO-1)	Count removed
Negative quantity	Drop rows with quantity < 1	Cannot interpret negative units sold	Count dropped
Negative price			
NULL customer_id			
Null-like invoice_date			
Future invoice_date			
Inaccurate total_revenue			

After cleaning, print a before/after summary:

- Row count before and after
- Null counts per column before and after
- Number of rows dropped per reason

> *Save the cleaned dataset as data/processed/retail_clean.csv before proceeding to Transformation.*

e. Transformation

Definition: Transformation restructures valid data to prepare it for the analytical model. Unlike Cleaning, this step runs unconditionally on all records that survived cleaning — it never drops rows.

Business objectives: BO-2, BO-3

Apply the following transformations:

- **Standardize country:** map all variants to title case using a lookup dict: {'colombia': 'Colombia', 'CO': 'Colombia', 'ecuador': 'Ecuador', …}
- **Parse invoice_date:** convert all surviving date strings to pandas datetime (handle YYYY/MM/DD and DD-MM-YYYY variants); extract year, month, day_of_week as separate columns
- **Cast customer_id:** convert float64 to Int64 (nullable integer) after sentinel fill
- **Normalize product:** strip whitespace and apply title case to ensure uniform casing
- **Feature engineering:** create revenue_bin column categorizing total_revenue into Low / Medium / High using quantile-based bins

> *Save the transformed dataset as data/processed/retail_transformed.csv. This file feeds both the post-transformation validation (Task f) and the dimensional model (Task g).*

f. Post-Transformation Validation (GE)

Business objectives: BO-1, BO-2, BO-3, BO-4

The purpose of this suite is different from Task b. Here, the Expectation Suite is a quality contract — all expectations must PASS. If any fail, the pipeline must not proceed to loading.
Reuse the same GE suite from Task b and run it against the transformed dataset. Additionally, add expectations that validate the new transformation-specific columns:

- invoice_date parsed as datetime type with no nulls
- month column in range [1, 12]
- country values only in {Colombia, Ecuador, Peru, Chile}
- revenue_bin only in {Low, Medium, High}
- total_revenue > 0 for all rows
- invoice_id unique with no nulls

Build a comparison table:

Expectation	Raw (pass %)	Clean (pass %)	Status	Dimension
expect_column_values_to_be_unique(invoice_id)	~77%	100%	RESOLVED	Uniqueness
expect_column_values_to_be_between(quantity, min=1)	~97%	100%	RESOLVED	Validity
expect_column_values_to_be_between(price, min=0.01)				
expect_column_values_to_be_in_set(country, …)				
Custom: total_revenue == qty × price				
expect_column_values_to_match_regex(date, YYYY-MM-DD)				
… (your additional expectations)	…	…	…	…

Compute and report:

- Data Quality Score (input) = passing expectations / total × 100
- Data Quality Score (output) = passing expectations / total × 100

Deliverables: Updated Expectation Suite JSON; Data Docs report for clean data; comparison table and DQ Scores in PDF.

g. Dimensional Modeling

Business objectives: BO-1, BO-2, BO-3
Design a star schema for the retail analytics scenario. The schema must support all KPIs in Task i.
Use the transformed dataset as the source for all tables.
Required Tables

Table	Type	Key Columns	Source Column(s)
dim_product	Dimension	product_id (surrogate PK), product_name	product

dim_customer	Dimension	customer_id (PK), customer_key	customer_id
dim_location	Dimension	location_id (surrogate PK), country	country
dim_date	Dimension	date_id (PK: YYYYMMDD int), full_date, year, month, month_name, day_of_week	invoice_date
fact_sales	Fact	sale_id (surrogate PK), invoice_id, product_id (FK), customer_id (FK), location_id (FK), date_id (FK), quantity, price, total_revenue	all columns

Granularity

One row in fact_sales = one transaction line (one invoice, one product, one customer, one date).

Design Requirements

- All foreign keys in fact_sales must reference valid primary keys in their dimension tables.
- dim_date must be fully generated for all dates in 2023 (not just dates present in the dataset).
- Surrogate keys must be integers, not the original source values.
- Include a model description document (reports/model_description.md) with an ERD diagram or table layout.

Deliverable: Star schema design diagram or description in PDF report; Python script that generates all dimension and fact DataFrames from the transformed dataset.

h. Load

Load all tables into a local SQLite database (data_warehouse.db) following the correct load order to maintain referential integrity:

1. Load dim_date first (no foreign keys)
2. Load dim_product, dim_customer, dim_location (no foreign keys)
3. Load fact_sales last (all FKs must resolve)

i. Business Analysis

Answer the question: "How are sales evolving over time, across products, customers, and regions?"

Note: KPIs and visualizations may be produced by querying the SQLite Data Warehouse or directly from the transformed DataFrame. Clearly state which approach you used.

Required KPIs and visualizations (one per Business Objective):

BO	KPI	Visualization	Notes
BO-1	Total revenue (sum of total_revenue)	Bar chart: total revenue per country	Exclude duplicate / negative rows — show impact of cleaning
BO-1	Average transaction value	Box plot: total_revenue distribution by product	Highlight outliers

BO-2	Monthly revenue trend	Line chart: monthly revenue Jan–Dec 2023	Use parsed month column from dim_date
BO-2	Peak day of week by transaction volume	Bar chart: transaction count by day of week	Use day_of_week from dim_date
BO-3	Top 3 products by total revenue	Horizontal bar chart: revenue by product	Sorted descending
BO-3	Sales distribution by country	Pie or choropleth: revenue share by country	Use standardized country names
BO-4	Data Quality Score before vs. after cleaning	Side-by-side bar: DQ score input vs. output	Values from Task f comparison table

All visualizations must be saved as PNG files in the reports/ folder.

> *Deliverable: visualization scripts in src/; PNG files in reports/; KPI values and chart screenshots in PDF report.*

8. Project Structure (suggested)

```
project/
├── data/
│   ├── raw/                  # retail_etl_dataset_enriched.csv (original, never modified)
│   └── processed/            # retail_clean.csv, retail_transformed.csv, data_warehouse.db
├── src/
│   ├── extract.py            # Task a: load + profiling
│   ├── validate_input.py     # Task b: GE input suite
│   ├── quality_analysis.py   # Task c: DQ issues table + policy proposal
│   ├── clean.py              # Task d: cleaning
│   ├── transform.py          # Task e: transformation
│   ├── validate_output.py    # Task f: GE output suite + comparison
│   ├── dimensional_model.py  # Task g: build dimension + fact tables
│   ├── load_dw.py            # Task h: load to SQLite DW
│   ├── analysis.py           # Task i: KPIs + visualizations
│   └── main.py               # Orchestrator: runs a → i in sequence
├── great_expectations/       # GE context, suites, validations
├── reports/
│   ├── quality_report.md     # Written DQ findings
│   ├── model_description.md  # Star schema ERD and table descriptions
│   └── *.png                 # Visualization outputs
├── requirements.txt
└── README.md                 # Setup and run instructions
```

9. Final Deliverables

a. PDF Report

A structured report containing all of the following sections in order:

1. Profiling summary (Task a)
2. Input validation results — failure rates per expectation (Task b)
3. Data Quality Issues table (Task c.1)
4. Data Quality Policy Proposal (Task c.2)
5. Cleaning Actions with justification and before/after summary (Task d)
6. Transformation steps applied and rationale (Task e)
7. Output validation results — comparison table + DQ Scores before and after (Task f)
8. Star schema design — ERD or table layout with column descriptions (Task g)
9. Referential integrity check results (Task h)
10. Visualizations and KPIs with interpretation (Task i)

b. Python Project (Git Repository)

11. All source scripts in src/ (extract.py through analysis.py + main.py)
12. Great Expectations configuration and Expectation Suite JSON files
13. Data Warehouse SQLite file (data/processed/data_warehouse.db)
14. Visualization PNG files (reports/*.png)
15. README.md with step-by-step instructions to reproduce the full pipeline
16. requirements.txt listing all dependencies with pinned versions